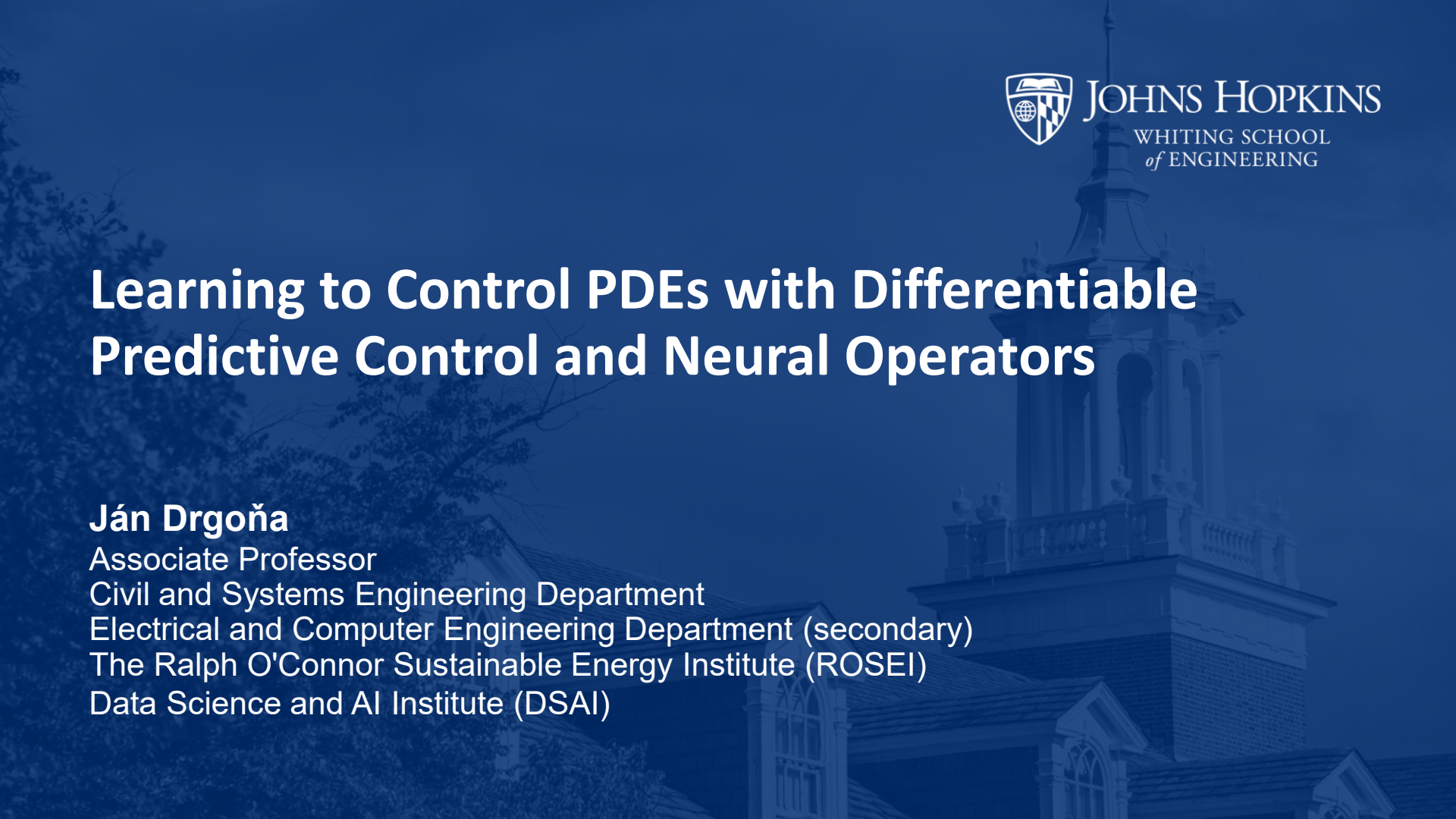


Learning to Control PDEs with Differentiable Predictive Control and Neural Operators

Ján Drgoňa
Associate Professor
Civil and Systems Engineering Department
Electrical and Computer Engineering Department (secondary)
The Ralph O'Connor Sustainable Energy Institute (ROSEI)
Data Science and AI Institute (DSAI)

State of the Art Control Methods

Model Predictive Control

95% of the industrial control applications are PID loops & expert driven rule-based control (RBC).
The rest is model predictive control (MPC).

Reinforcement Learning

Everywhere where you have reliable digital environment model, and you can use compute resources to solve the problem offline. Successful in GenAI, games, and robotics.

James B. Rawlings, David Q. Mayne, Moritz M. Diehl, Model Predictive Control: Theory and Design, Nob Hill Publishing, 2nd ed., 2017
Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction, MIT Press, 2nd ed., 2018
Drgona J., et al. 2020. "All You Need to Know About Model Predictive Control for Buildings." Annual Reviews in Control, September 29, 2020
G. Dulac-Arnold, D. J. Mankowitz, and T. Hester, "Challenges of real-world reinforcement learning,", 2019.

Model Predictive Control (MPC)

MPC uses a **receding horizon** approach to compute the optimal control input at each time step by solving a **finite-horizon** optimal control problem in real-time via **online optimization** using numerical methods.

Finite-horizon optimal control problem formulation:

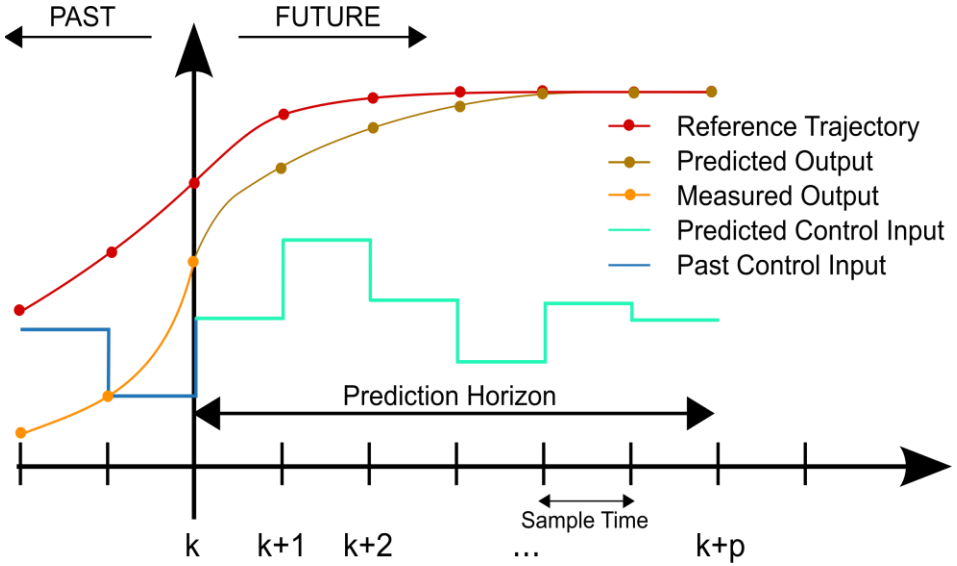
$$\min_{\mathbf{u}_0,\ldots,\mathbf{u}_{N-1}} \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k) + p_N(\mathbf{x}_N)$$

$$\text{s.t. } \mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k),\ k \in \mathbb{N}_0^{N-1}$$

$$h(\mathbf{x}_k) \leq 0$$

$$g(\mathbf{u}_k) \leq 0$$

$$\mathbf{x}_0 = \mathbf{x}(t)$$



At each time step:

1. Measure the current state x_{current}.
2. Solve the finite-horizon optimization problem for N future steps.
3. Apply only the first control input u_0^*.
4. Move to the next time step and repeat.

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MPC software tools:

PYOMO, MATLAB, JUMP, GEKKO DYNAMIC OPTIMIZATION, CVXPY, CasADi, GUROBI OPTIMIZATION, robust optimal control toolbox mpc

Challenges: Computationally expensive online.

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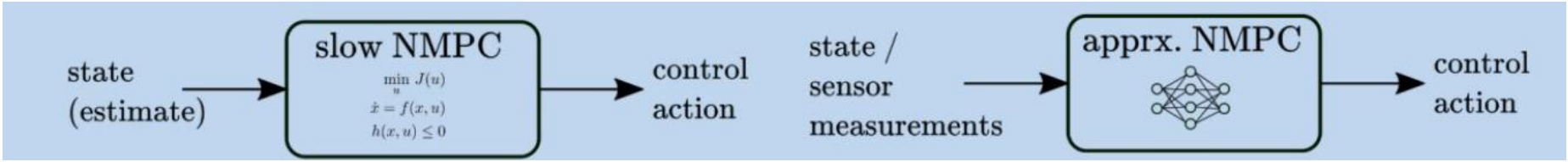
Approximate MPC



Step 1: solve set of MPC problems to generate labeled training data

$$\min_{\mathbf{u}_0,\ldots,\mathbf{u}_{N-1}} \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k) + p_N(\mathbf{x}_N)$$

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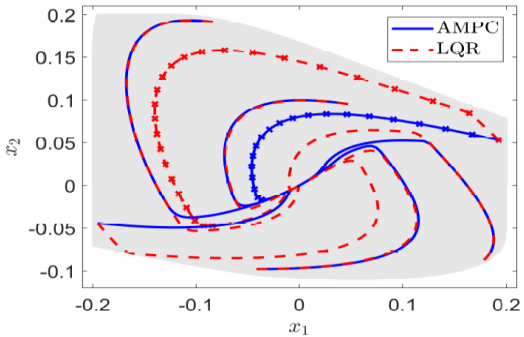
Step 2: supervised imitation learning to learn approximate MPC policy

Machine Learning

Sample robust MPC

Learn: $\pi_{\text{approx}} \approx \pi_{\text{MPC}}$



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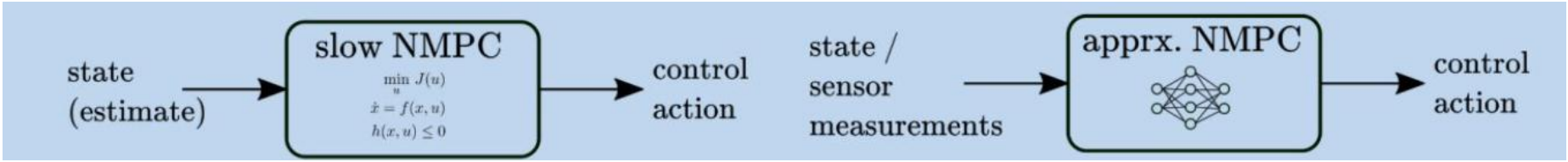

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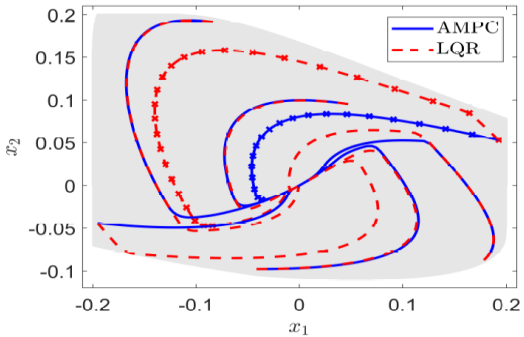

Challenges:

Expensive training data generation.

Knowledge of the model and constraints not utilized in learning.

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Reinforcement Learning (RL)

RL is a **data-driven decision-making framework** where an agent interacts with an environment to learn an optimal policy that maximizes rewards over **infinite-horizon**. Most modern **policy-gradient methods** (actor–critic, PPO, TRPO, DDPG, TD3, offline RL) **learn a critic** and use it to **update the actor (policy)**.

1, **Data collection** via sampling:

$$
\begin{aligned}
&x_t \sim \text{environment} \\
&u_t = \pi_\theta(x_t) \\
&x_{t+1},\ r_t \sim \mathcal{P}(\cdot \mid x_t, u_t) \\
&(x_t,\ u_t,\ r_t,\ x_{t+1}) \in \mathcal{D}
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$$

2, **Critic Update:**

Q-function

$$Q^\pi(x,u) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k r_k \mid x_0 = x, u_0 = u, \pi\right]$$

Bellman target

$$y_t = r_t + \gamma Q_{\phi'}(x_{t+1},\ \pi_{\theta'}(x_{t+1}))$$

Critic loss function

$$\mathcal{L}_{\text{critic}}(\phi) = \left(Q_\phi(x_t, u_t) - y_t\right)^2$$

3, **Actor update** via deterministic policy gradient:

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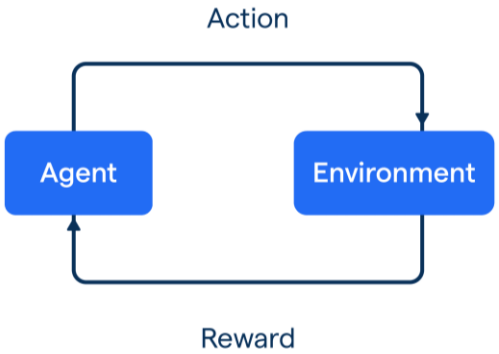

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RL software tools:

PyTorch TensorFlow

Gym

Challenges: Sample inefficient! Hard to obtain guarantees.

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Model-Based Reinforcement Learning (MBRL)

MBRL aims to **improve sample efficiency** by **learning a predictive model** of the environment dynamics and using it to **simulate trajectories** for Q-function fitting and policy optimization. This framework is followed by methods such as MBPO, TD3-MB, MOPO, COMBO, DreamerV3, and Plan2Explore.

1, **Model rollouts** via sampling:

$$\hat{x}_0 = x \sim \mathcal{D}$$
$$\hat{u}_t \sim \pi_\theta(\cdot \mid \hat{x}_t)$$
$$\hat{x}_{t+1} = f_\psi(\hat{x}_t, \hat{u}_t)$$
$$\hat{r}_t = \hat{r}_\psi(\hat{x}_t, \hat{u}_t)$$

2, **Critic Update** using model rollouts:

Bellman target $y_t = \hat{r}_t + \gamma Q_{\phi'}(\hat{x}_{t+1}, \pi_{\theta'}(\hat{x}_{t+1}))$

Critic loss function $\mathcal{L}_{\text{critic}}(\phi) = (Q_\phi(\hat{x}_t, \hat{u}_t) - y_t)^2$

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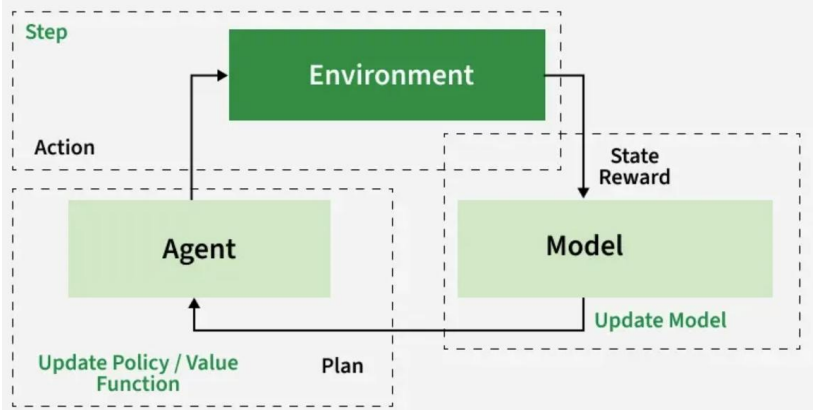

Image: www.geeksforgeeks.org/artificial-intelligence/model-based-reinforcement-learning-mbrl-in-ai/

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Critic loss function $\mathcal{L}_{\text{critic}}(\phi) = (Q_\phi(\hat{x}_t, \hat{u}_t) - y_t)^2$

Why to learn the critic if we have accurate and differentiable model and parametric reward functions?

3, **Actor update** via deterministic policy gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\hat{x}_t \sim \mathcal{D}} \left[\nabla_u Q_\phi(\hat{x}_t, u)\big|_{u=\pi_\theta(\hat{x}_t)} \cdot \nabla_\theta \pi_\theta(\hat{x}_t) \right]$$

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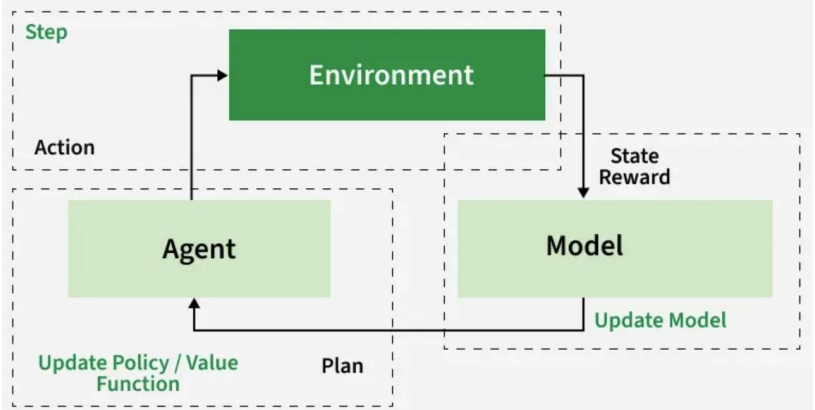

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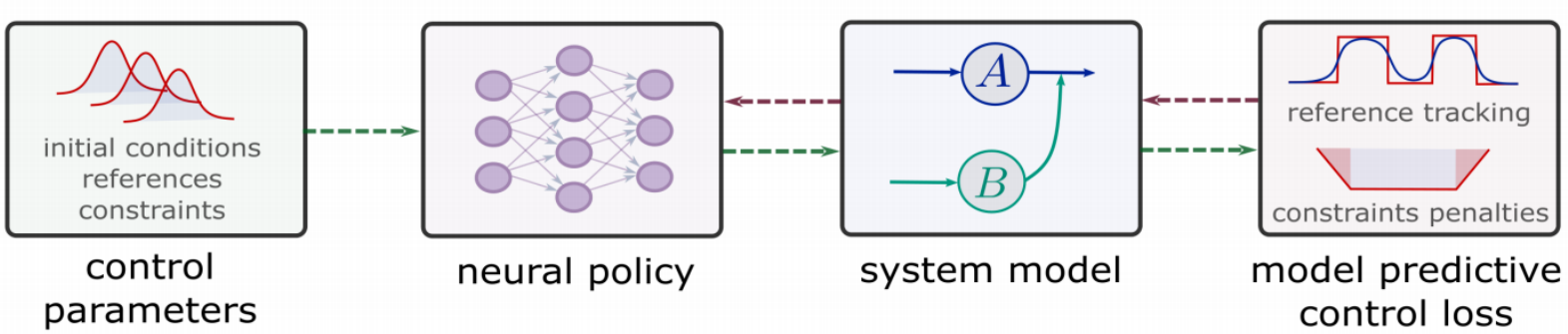
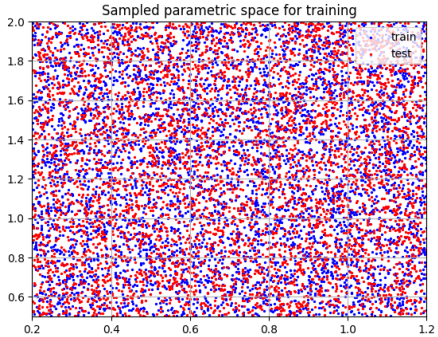
Differentiable Predictive Control (DPC)



Dataset: collocation points in the control parametric space.



Architecture: differentiable model with neural network control policy.

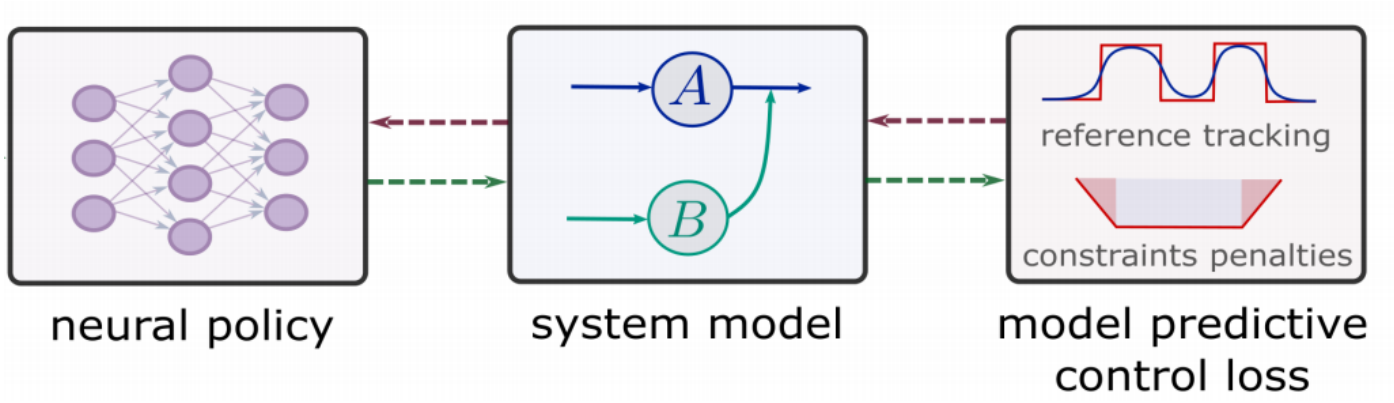
$$x_{k+1} = \text{ODESolve}(f(x_k, u_k))$$
$$u_k = NN_\theta(x_k, \xi_k)$$
$$g(x_k, u_k, \xi_k) \leq 0$$
$$x_0 \sim \mathcal{P}_{x_0}$$
$$\xi_k \sim \mathcal{P}_\xi$$

Loss function: reference tracking, constraints and terminal penalties.

$$\ell_1 = \sum_{i=1}^{m} \sum_{k=1}^{N-1} Q_x ||x_k^i - r_k^i||_2^2$$
$$\ell_2 = \sum_{i=1}^{m} \sum_{k=1}^{N-1} Q_g ||\texttt{RELU}(g(x_k^i, u_k^i, \xi_k^i)||_2^2$$
$$\ell_{L2C} = \ell_1 + \ell_2$$

J. Drgoňa, A. Tuor and D. Vrabie, "Learning Constrained Parametric Differentiable Predictive Control Policies With Guarantees," in IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024
Ján Drgoňa, et al, Differentiable predictive control: Deep learning alternative to explicit model predictive control for unknown nonlinear systems, Journal of Process Control, 2022

Differentiable Closed-Loop System



Forward pass. For a single scenario, the closed-loop recursion and loss are

$$
\begin{aligned}
u_k &= \pi_\theta(x_k;\xi_k),\\
x_{k+1} &= f(x_k,u_k;\xi_k),\\
\ell_k &= \ell(x_k,u_k;\xi_k) \;+\; p_x\big(g(x_k;\xi_k)\big) \;+\; p_u\big(h(u_k;\xi_k)\big),\\
\mathcal{L} &= \frac{1}{N}\sum_{k=0}^{N-1}\ell_k \;+\; \ell_N(x_N;\xi_N).
\end{aligned}
$$

Differentiable Closed-Loop System

Backward pass (sensitivities). Let

$$A_k := \frac{\partial f}{\partial x}, \quad B_k := \frac{\partial f}{\partial u}, \quad P_k := \frac{\partial \pi_\theta}{\partial x}, \quad G_k := \frac{\partial \pi_\theta}{\partial \theta} \quad \text{(all evaluated at } (x_k, u_k; \xi_k)).$$

Define $s_N := \frac{\partial \ell_N}{\partial x}(x_N; \xi_N)$ and stage gradients

$$\ell_{x,k} := \frac{\partial \ell}{\partial x} + \frac{\partial p_x}{\partial x}, \qquad \ell_{u,k} := \frac{\partial \ell}{\partial u} + \frac{\partial p_u}{\partial u}.$$

Then the reverse recursion for $k = N-1, \ldots, 0$ is

$$s_k \;=\; \ell_{x,k} \;+\; A_k^\top s_{k+1} \;+\; P_k^\top \big(\ell_{u,k} + B_k^\top s_{k+1}\big),$$

and the gradient w.r.t. the policy parameters is

$$\nabla_\theta \mathcal{L} \;=\; \sum_{k=0}^{N-1} G_k^\top \big(\ell_{u,k} + B_k^\top s_{k+1}\big).$$

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Backward pass in DPC are discrete adjoint sensitivities leading to Pontryagin costates in continuous time.

DPC Policy Optimization Algorithm

Algorithm 5: Self-Supervised DPC Policy Optimization

Input: Horizon N, batch size B; initial-state distribution $\mathcal{X}_0$, scenario distribution $\mathcal{D}$; dynamics $x_{k+1} = f(x_k, u_k; \xi_k)$; policy class $u_k = \pi_\theta(x_k; \xi_k)$; costs ℓ, ℓ_N; penalties p_x, p_u with weights Q_x, Q_u; constraints g, h; optimizer $\mathbb{O}$

Output: Trained explicit policy $\pi_{\theta^\star}$

while *not converged* **do**

  Sample mini-batch $\{(x_0^{(i)}, \xi^{(i)})\}_{i=1}^{B} \sim \mathcal{X}_0 \times \mathcal{D}$;

  for $i \leftarrow 1$ **to** B **do**

    for $k \leftarrow 0$ **to** $N-1$ **do**

      $u_k^{(i)} \leftarrow \pi_\theta\big(x_k^{(i)}; \xi_k^{(i)}\big)$;

      $x_{k+1}^{(i)} \leftarrow f\big(x_k^{(i)}, u_k^{(i)}; \xi_k^{(i)}\big)$;

      $\ell_k^{(i)} \leftarrow \ell\big(x_k^{(i)}, u_k^{(i)}; \xi_k^{(i)}\big) + Q_x\, p_x\big(g(x_k^{(i)}; \xi_k^{(i)})\big) + Q_u\, p_u\big(h(u_k^{(i)}; \xi_k^{(i)})\big)$;

    $\mathcal{L}^{(i)} \leftarrow \frac{1}{N} \sum_{k=0}^{N-1} \ell_k^{(i)} + \ell_N\big(x_N^{(i)}; \xi_N^{(i)}\big)$;

  $\mathcal{L} \leftarrow \frac{1}{B} \sum_{i=1}^{B} \mathcal{L}^{(i)}$; // batch DPC loss

  Compute $\nabla_\theta \mathcal{L}$ (e.g., BPTT or adjoint sensitivity using (8.3)–(8.4));

  $\theta \leftarrow \mathbb{O}\big(\theta, \nabla_\theta \mathcal{L}\big)$; // e.g., Adam step

return π_θ;

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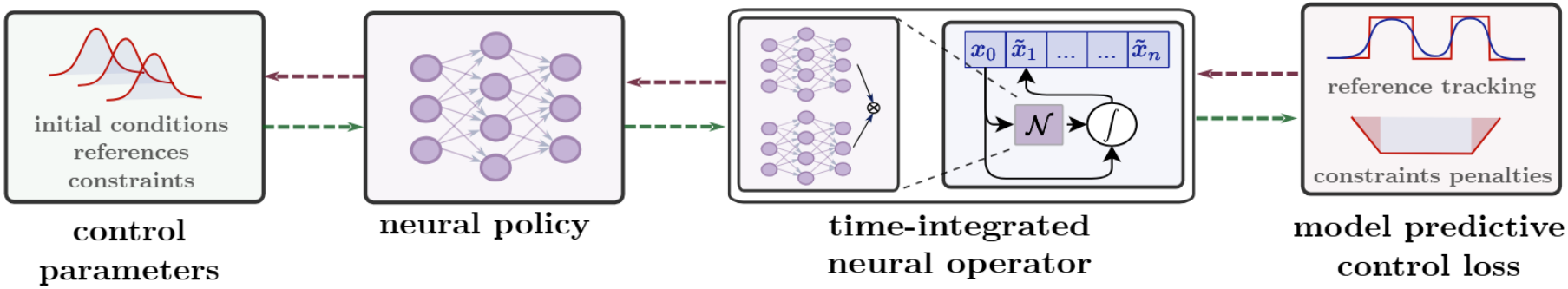
Ok, but what if the system is PDE?

Learning to Control PDEs with DPC and Neural Operators



$$\min_{a(\cdot,\cdot)} \quad J[u,a] = \int_0^T \int_\Omega \ell(u(t,\mathbf{x}), a(t,\mathbf{x}), \boldsymbol{\xi}(t))\, d\mathbf{x}\, dt + \int_\Omega \ell_T(u(T,\mathbf{x}))\, d\mathbf{x},$$

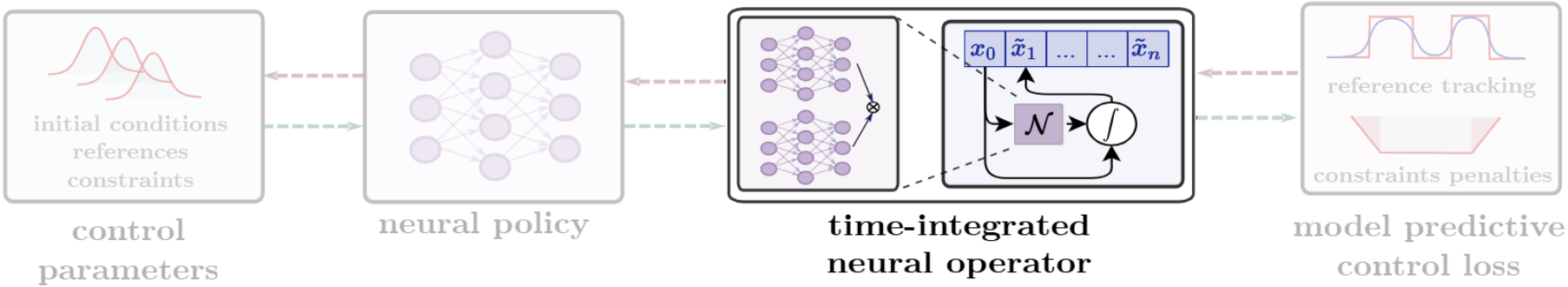
$$\text{s.t.} \quad \frac{\partial u}{\partial t} = \mathcal{F}(t, \mathbf{x}, u, \nabla u, \nabla^2 u, \ldots, a),$$

$$h(u(\mathbf{x},t), \boldsymbol{\xi}(t)) \leq 0, \quad g(a(\mathbf{x},t), \boldsymbol{\xi}(t)) \leq 0,$$

$$u(\mathbf{x},0) = u_0(\mathbf{x}), \quad \mathcal{B}(u) = 0.$$

D. R. Sharkar, J. Drgoňa, S. Goswami, "Learning to Control PDEs with Differentiable Predictive Control and Time-Integrated Neural Operators," arXiv:2511.08992, 2025

Time-Integrated Neural Operator (TI-DeepOnet)



TI-DeepOnet: PDE surrogate based on method-of-lines spatial discretization

$$\partial u/\partial t \approx \mathcal{G}_\theta(u^i, a^i)(\zeta) = \sum_{j=1}^{p} \left[b_j^u(u^i) \odot b_j^a(a^i) \right] t_j(\zeta)$$

$\odot$ denotes element-wise multiplication

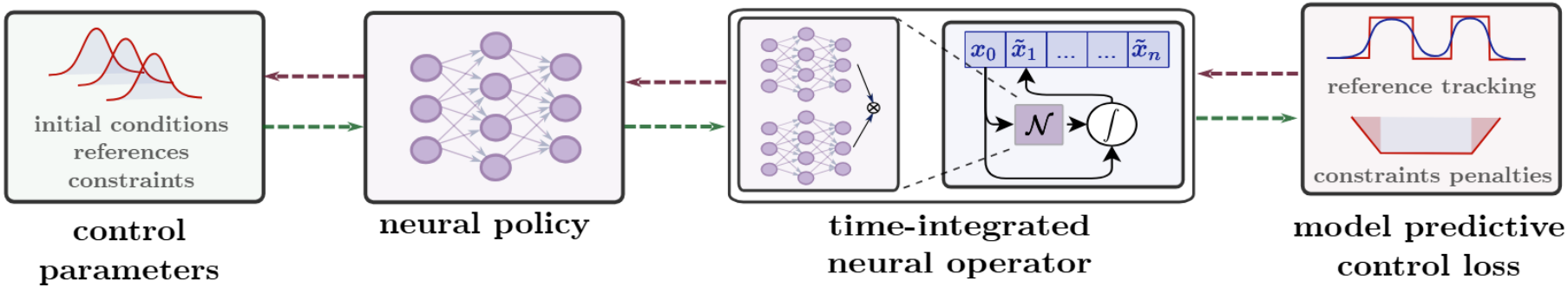
b_j^u is a state branch net encoding the state field

b_j^a is a control branch net encoding the control function

t_j is a trunk net encoding the solution at the collocation points ζ

D. R. Sharkar, J. Drgoňa, S. Goswami, "Learning to Control PDEs with Differentiable Predictive Control and Time-Integrated Neural Operators," arXiv:2511.08992, 2025

Formulation of DPC with TI-DeepOnet



Discretize then optimize reformulation of the PDE control problem:

$$\min_{\mathbf{W}} \quad \mathbb{E}_{u_0 \sim \mathcal{P}_{u_0}, \boldsymbol{\xi} \sim \mathcal{P}_{\boldsymbol{\xi}}} \left[\sum_{k=0}^{N-1} \sum_{i=1}^{n_x} \ell(u_k^i, a_k^i, \boldsymbol{\xi}_k) \Delta x + \sum_{i=1}^{n_x} \ell_N(u_N^i) \Delta x \right],$$

$$\text{s.t.} \quad \mathbf{u}_{k+1} = \text{ODESolve}(\mathcal{G}_{\boldsymbol{\theta}}(t_k, \mathbf{u}_k, \mathbf{a}_k)),$$

$$\mathbf{a}_k = \pi_{\mathbf{W}}(\mathbf{u}_k, \boldsymbol{\xi}_k), \quad h(\mathbf{u}_k, \boldsymbol{\xi}_k) \leq 0, \quad g(\mathbf{a}_k, \boldsymbol{\xi}_k) \leq 0,$$

D. R. Sharkar, J. Drgoňa, S. Goswami, "Learning to Control PDEs with Differentiable Predictive Control and Time-Integrated Neural Operators," arXiv:2511.08992, 2025

Training Algorithms for TI-DeepOnet and DPC

Algorithm 1 Train TI-DON

1: Training datasets of initial condition, problem parameters ξ_k and control inputs from $\mathcal{P}_{u_0}$, $\mathcal{P}_\xi$ and $\mathcal{P}_a$ respectively.
2: Neural operator, $\mathcal{G}_\theta : (u(t), f(t)) \mapsto \partial_t u$
3: Time stepping, $u(t+1) = \text{RK4}(\mathcal{G}_\theta)$
4: Mean square error loss, $\mathcal{L}_{\text{mse}}$
5: Optimizer $\mathbb{O}$
6: Learn, θ via optimizer $\mathbb{O}$ using gradient $\nabla_\theta \mathcal{L}_{\text{mse}}$
7: **return** trained neural operator, $\mathcal{G}_\theta$

Algorithm 2 DPC offline training

1: Training datasets of initial condition and problem parameters ξ_k from $\mathcal{P}_{u_0}$ and $\mathcal{P}_\xi$ respectively.
2: Pretrained neural operator, $\mathcal{G}_\theta$
3: Time stepping, $u(t+1) = \text{RK4}(\mathcal{G}_\theta)$
4: DPC loss. $\mathcal{L}_{\text{DPC}}$ (6)
5: Optimizer $\mathbb{O}$
6: Learn, $\mathbf{W}$ via optimizer $\mathbb{O}$ using gradient $\nabla_{\mathbf{W}} \mathcal{L}_{\text{DPC}}$
7: **return** optimized policy $\pi_{\mathbf{W}}(\mathbf{u}_k, \boldsymbol{\xi}_k)$

JAX implementation: https://github.com/Centrum-IntelliPhysics/PDEControl_DPC

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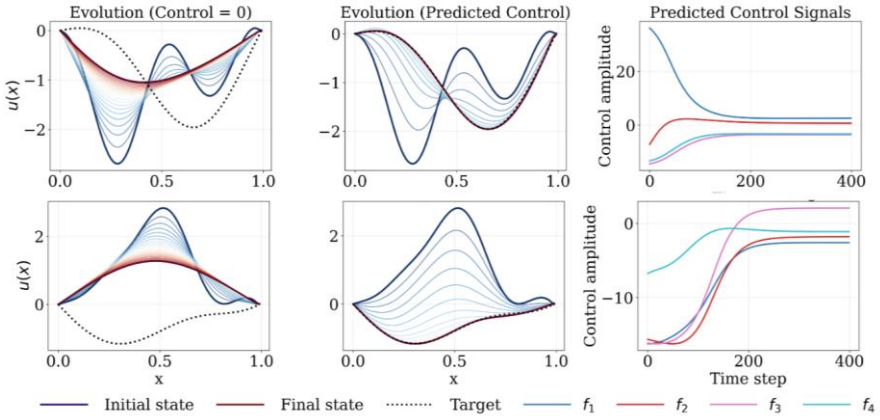
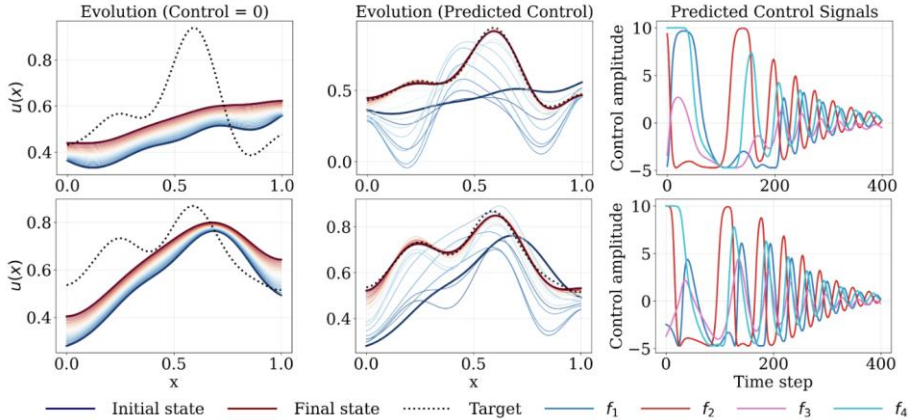
Learning to Control PDEs with DPC and Neural Operators

Algorithm	Metric	HE TT (IV-B)	BE (IV-C)	RDE (IV-D)
NMPC	MSE	$(4.2 \pm 6.7) \times 10^{-3}$	$(3.1 \pm 2.9) \times 10^{+6}$	$(1.8 \pm 1.6) \times 10^{-3}$
	Time/step (s)	4.3 ± 1.5	3.9 ± 13	7.9 ± 1.6
TIDON-DPC	MSE	$(7.7 \pm 20) \times 10^{-3}$	$(4.4 \pm 9.3) \times 10^{+7}$	$(1.4 \pm 1.4) \times 10^{-3}$
	Time/step (s)	$(2.5 \pm 2.2) \times 10^{-4}$	$(8.8 \pm 1.3) \times 10^{-5}$	$(2.4 \pm 2.4) \times 10^{-4}$

Heat equation tracking control

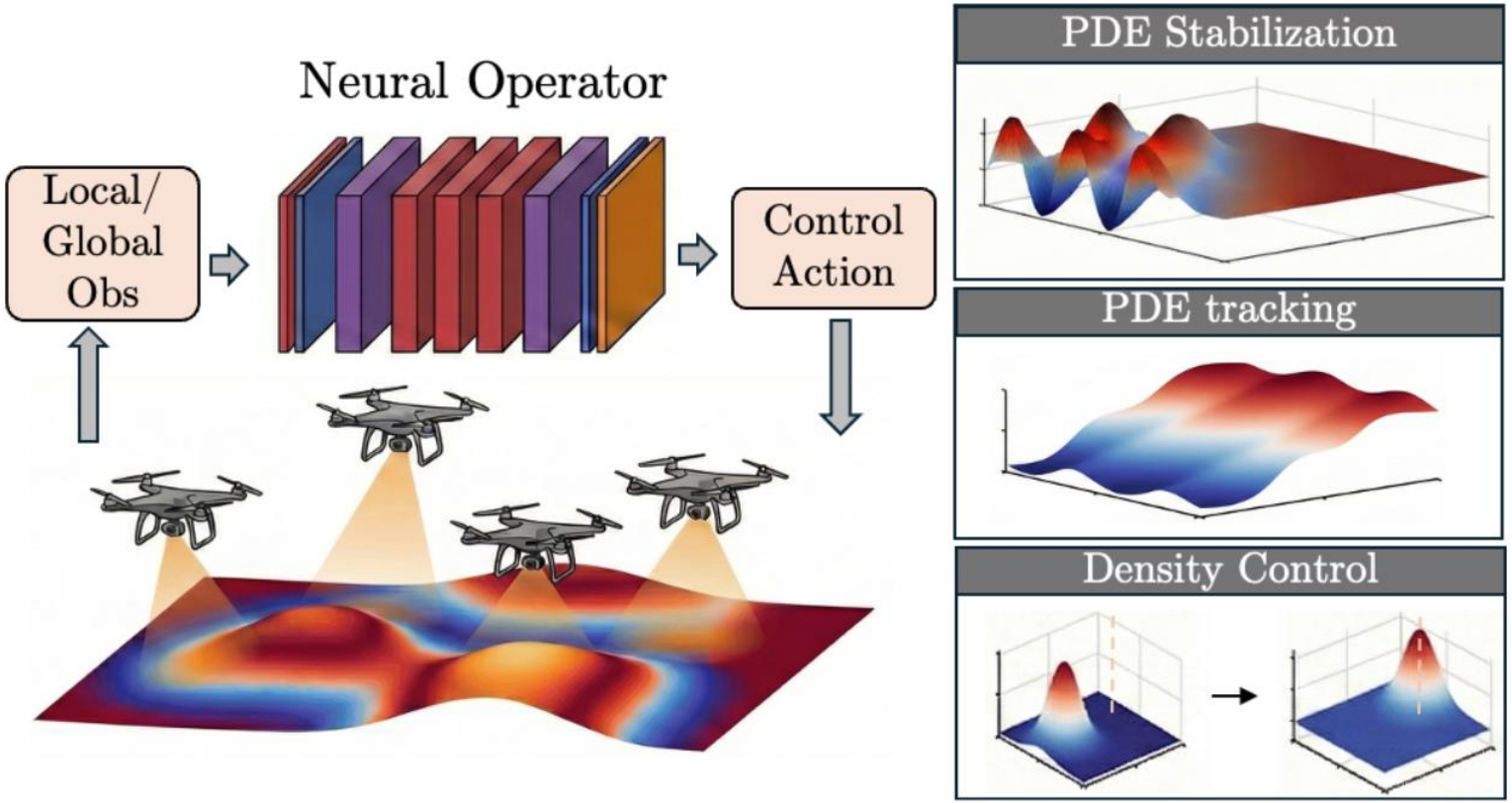


Fisher-KPP population density control



D. R. Sharkar, J. Drgoňa, S. Goswami, "Learning to Control PDEs with Differentiable Predictive Control and Time-Integrated Neural Operators," arXiv:2511.08992, 2025

Cardinality-Invariant Neural Operator Policies for Multi-Agent PDE Control



P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Multi-agent PDE Control Problem

$$\min_{\boldsymbol{\theta}} \quad \mathcal{J} = \mathbb{E}_{\boldsymbol{\psi}\sim\mathcal{D}_{\text{prob}}}\left[\int_0^T \ell\big(z(\boldsymbol{x},t),\boldsymbol{u}^M(t),\boldsymbol{\psi}\big),dt\right] \longrightarrow \text{Parametric control functional}$$

$$\text{s.t.} \quad \frac{\partial z}{\partial t} = \mathcal{F}\left(z, \sum_{i=1}^{M} \mathcal{B}(\boldsymbol{x}, u_i(t), \boldsymbol{\xi}_i(t))\right), \longrightarrow \text{Time-dependent PDE operator}$$

$$\dot{\boldsymbol{\xi}}_i(t) = \mathcal{K}(\boldsymbol{\xi}_i(t), \boldsymbol{v}_i(t)), \quad i = 1,\dots,M, \longrightarrow \text{Actuator ODE dynamics}$$

$$z_i^{\text{patch}}(t) = \mathcal{O}bs(z(\boldsymbol{x},t), \boldsymbol{\xi}_i(t), \boldsymbol{\psi}), \longrightarrow \text{Sensor operator}$$

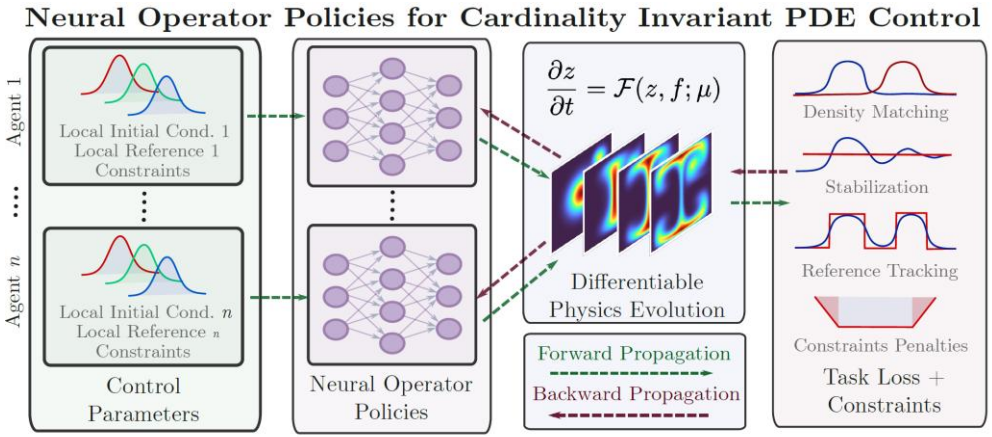
$$[u_i(t), \boldsymbol{v}_i]^\top = \mathcal{G}_{\boldsymbol{\theta}}(z_i^{\text{patch}}, \boldsymbol{\xi}_i), \longrightarrow \text{Shared neural operator policy}$$

$$\boldsymbol{c}(z(\boldsymbol{x},t), \boldsymbol{u}^M(t), \boldsymbol{\psi}) \le 0. \longrightarrow \text{Constraints}$$

P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

PDE Control as Self-Supervised Neural Operator Learning

We learn the multi-agent policy operator using differentiable predictive control algorithm with differentiable PDE solvers.

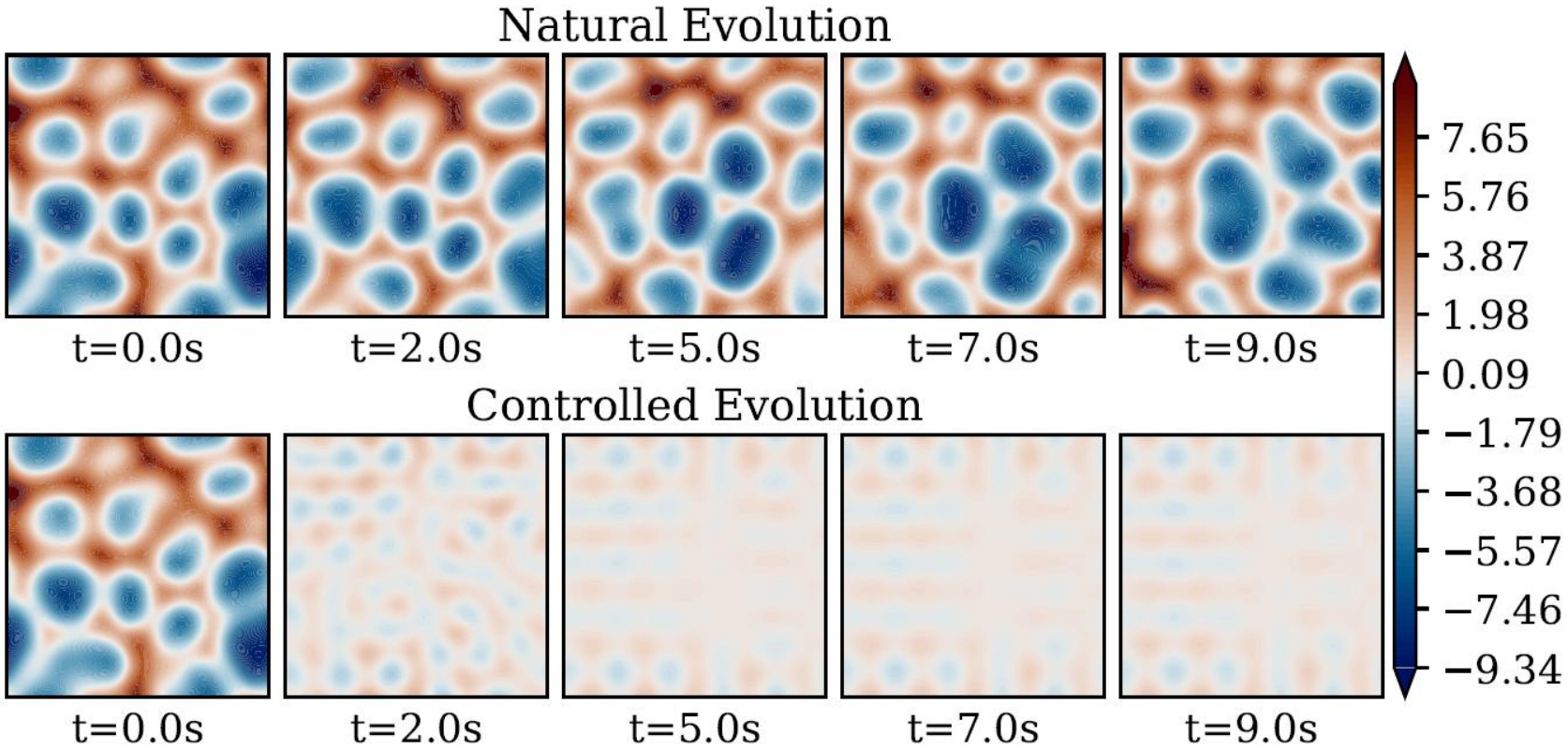


Algorithm 1 Self-Supervised Operator Policy Training

Input: distribution $\mathcal{D}_{\text{prob}}$, horizon T, number of agents M
Initialize: shared policy parameters $\boldsymbol{\theta}$
while not converged **do**
  {**1. Data Sampling**}
  Sample instances $\boldsymbol{\psi} = (z_0, z_{\text{target}}, \dots) \sim \mathcal{D}_{\text{prob}}$
  Initialize state z_0, actuator positions $\{\boldsymbol{\xi}_{i,t=0}\}_{i=1}^{M}$
  for $t = 0$ to T **do**
    {**2. Shared Policy Inference**}
    for $i = 1$ to M **do**
      $z_i^{\text{patch}} \leftarrow \mathcal{O}bs(z_t, \boldsymbol{\xi}_{i,t}, \boldsymbol{\psi})$
      $[u_i(t), \boldsymbol{v}_i]^\top \leftarrow \mathcal{G}_{\boldsymbol{\theta}}(z_i^{\text{patch}}, \boldsymbol{\xi}_{i,t})$
    end for
    {**3. Differentiable Physics Step**}
    $z_{t+1}, \{\boldsymbol{\xi}_{t+1}\} \leftarrow \Psi(z_t, \{\boldsymbol{\xi}_t\}, \boldsymbol{u}_t^M)$
    {**4. Loss**}
    $\mathcal{J} \leftarrow \mathcal{J} + \ell(z_{t+1}, \boldsymbol{u}_t^M, \boldsymbol{\psi})$
  end for
  Update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{J}$ {Backprop via Physics}
end while

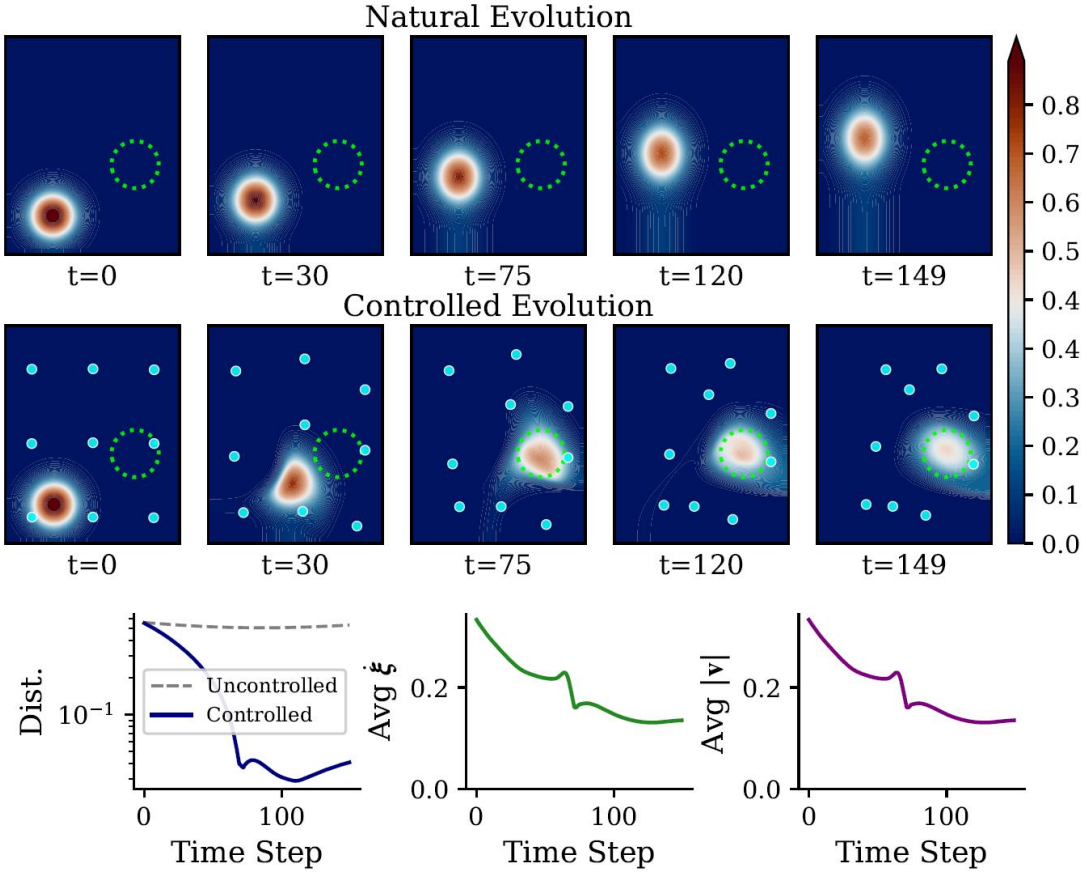
P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Empirical Results: Stabilization of 2D Kuramoto-Sivashinsky Chaos



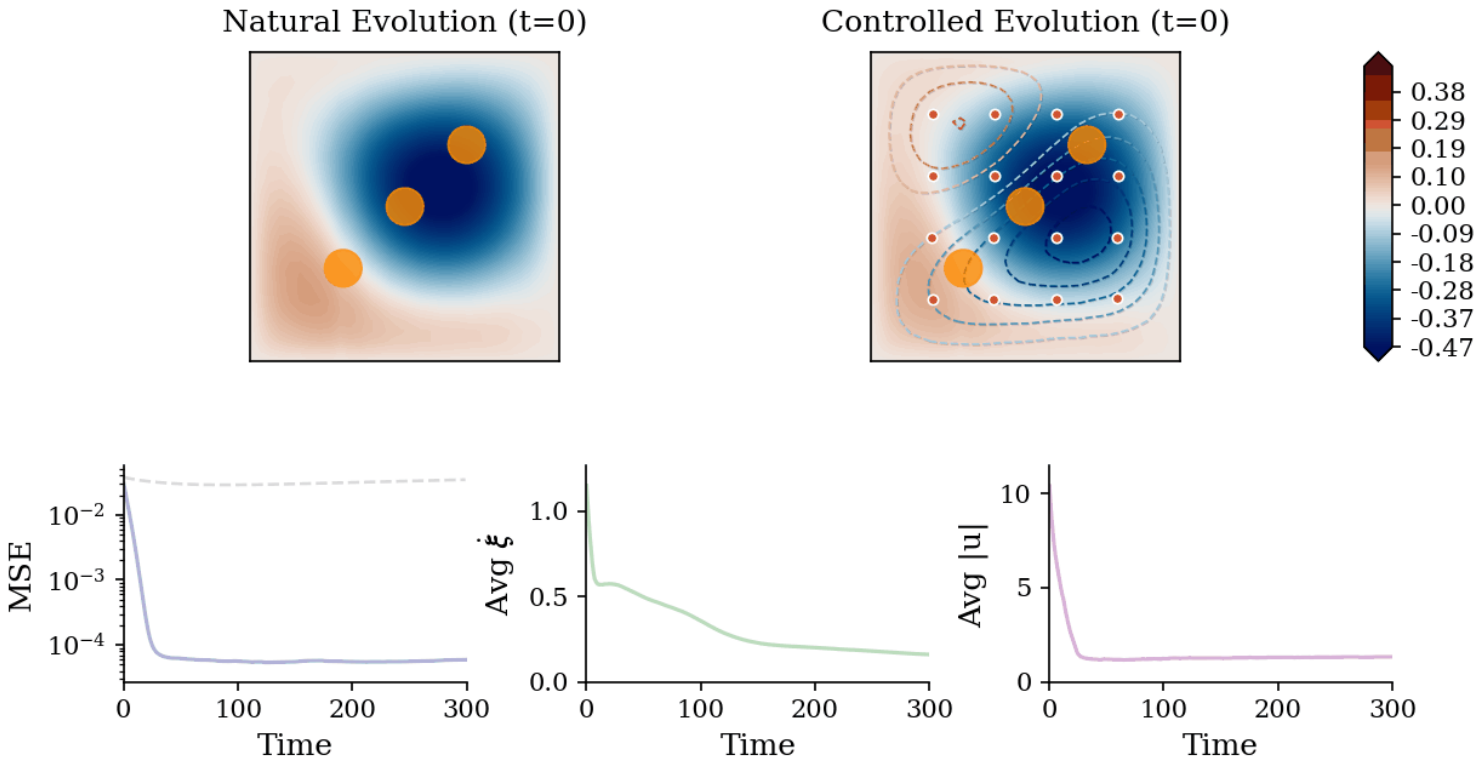
P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Empirical Results: Density Transport in Advection-Diffusion



P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, under review, 2026, https://github.com/SOLARIS-JHU/Multi-Agent-DPC

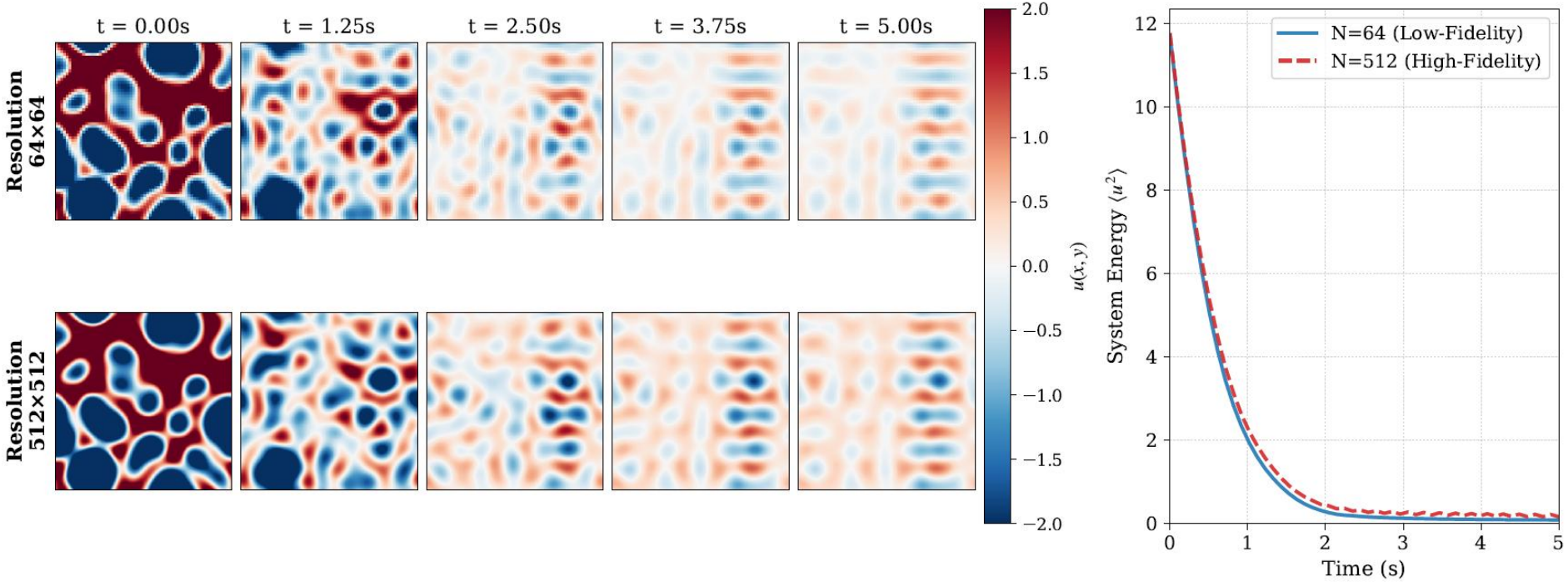
Empirical Results: Multi-agent Heat Control with Spatial Constraints



P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Empirical Results: Solver Fidelity Robustness

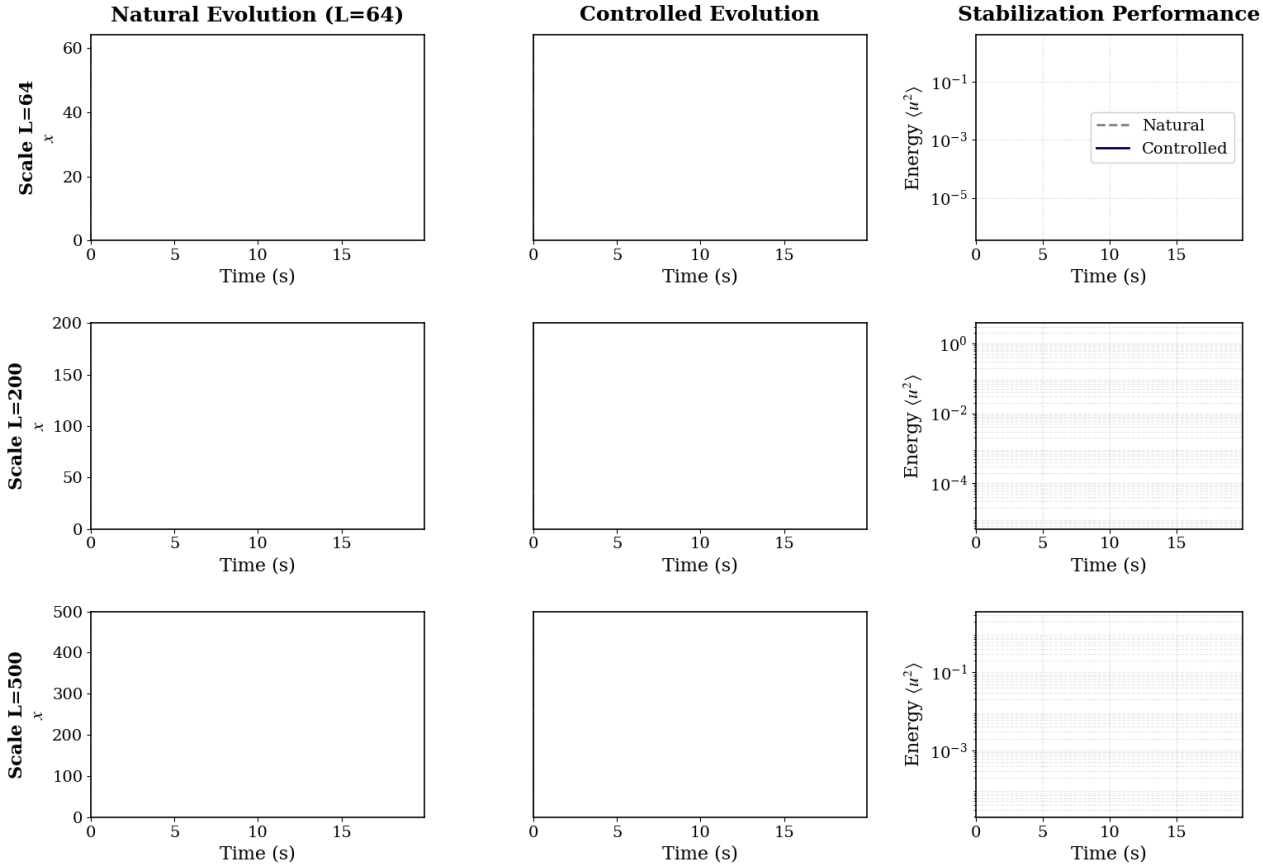
The policy, trained on a coarse discretization of 64 × 64 (Top), is deployed zero-shot on a high-fidelity solver with 512 × 512 grid points (Bottom).



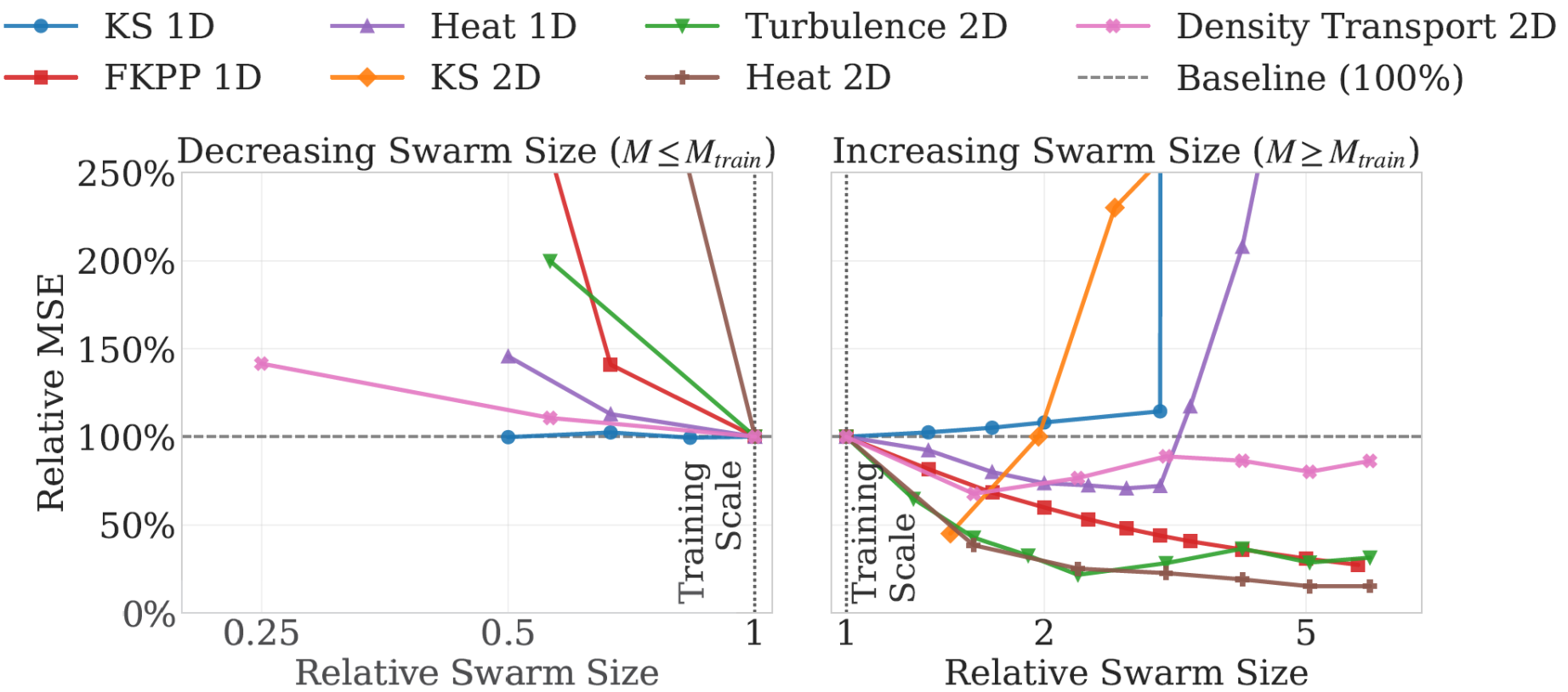
P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Empirical Results: Solver Fidelity Robustness

Stabilizing 1D Kuramoto-Sivashinsky across actuator scales with single neural operator policy.



Cardinality Invariance Across Diverse PDE Benchmarks



P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Mean-Field Gradient Consistency

We train a **shared neural operator policy** on a *finite swarm* of size M, but **deploy it zero-shot** on *different swarm sizes*. Key questions:

- Does finite-M training optimize the *true continuous control problem*?
- Why does the policy transfer across cardinalities without knowing M?

We observe that training on a finite swarm asymptotically optimizes the *continuous PDE control operator*, not a particular agent configuration. Hence cardinality becomes a *resolution parameter.*

As the number of agents grows, discrete actuation converges to a **continuous actuator density**.

$$\sum_{i=1}^{M} B(x, \xi_i) u_i \longrightarrow \mathcal{B}_\infty(z) \quad \text{(mean-field forcing operator)}$$

Our primary theoretical result establishes that the gradients computed during our discrete multi-agent training are consistent estimators of the gradients for this continuous limit.

P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

Mean-Field Gradient Consistency

Theorem 5.1 (Consistency of Discrete Policy Gradients (Informal)). *As the number of agents $M \to \infty$, the discrete policy gradient $\nabla_{\boldsymbol{\theta}} \mathcal{J}_M$ converges to the mean-field gradient $\mathcal{D}_{\boldsymbol{\theta}} \mathcal{J}_\infty$.*

Proof sketch: Proof relies on the discrete physics uniformly approximates the mean-field physics.

1. **Uniform bounds:** PDE states and adjoints remain uniformly bounded in suitable function space regardless of *M*, preventing blow-ups as agents are added.
2. **Compactness and convergence:** Using the Aubin-Lions lemma, we extract strongly convergent subsequences of the state trajectories, showing that the discrete dynamics converge to a unique mean-field limit.
3. **Gradient Identification:** We decompose the gradient error into a state-dependent term and a measure-dependent term. We show both vanish in the limit, proving that optimizing the swarm is asymptotically equivalent to optimizing the continuous operator.

P. Zanotta, H. Zheng, D. R. Sharkar, S. Goswami, J. Drgoňa, Cardinality-Invariant Neural Operator Policies for Scalable PDE Control, ICML, 2026, https://github.com/SOLARIS-JHU/CINOC

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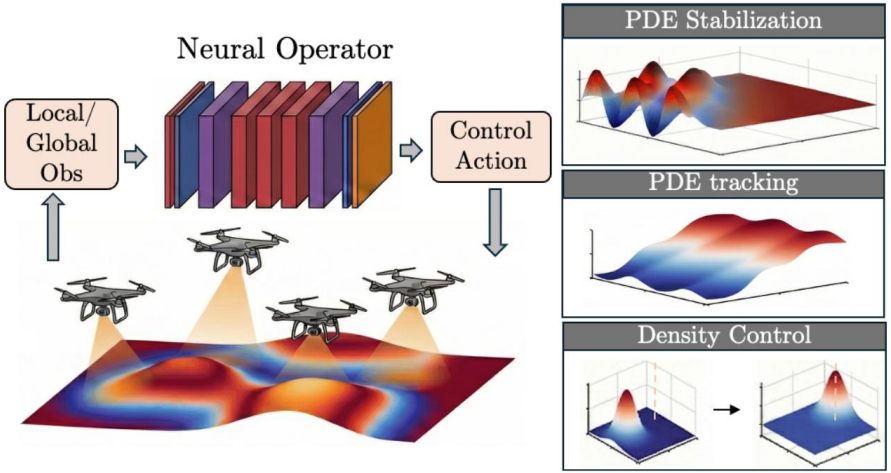
U.S. DEPARTMENT OF ENERGY

Summary

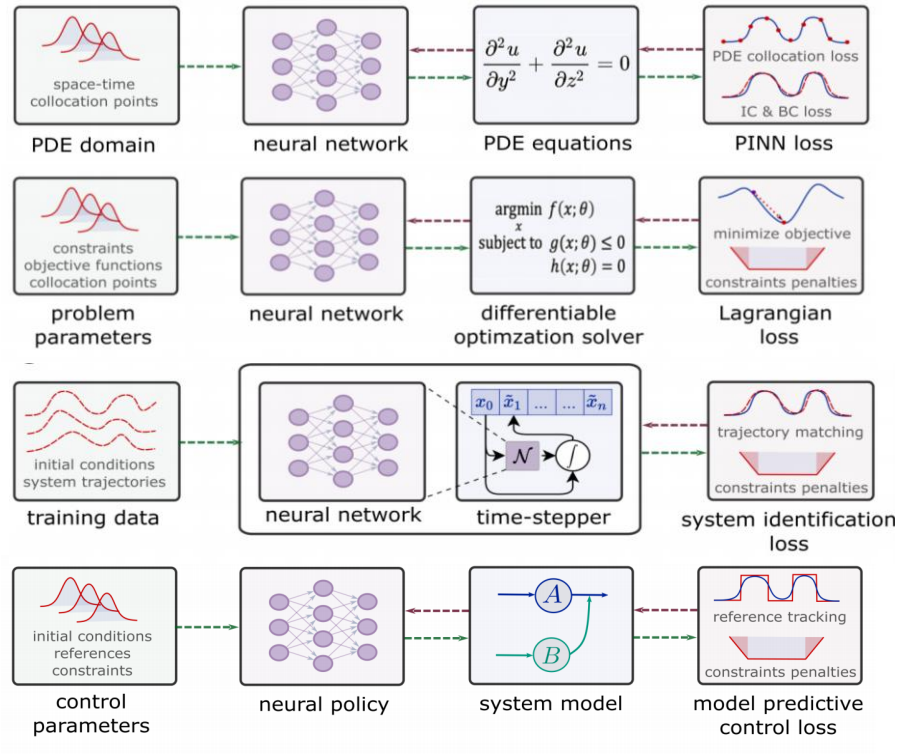
- **Differentiable Predictive Control (DPC) with Neural Operators**
  - Enables scalable policy optimization for PDEs
  - Works both with differentiable PDE solves and neural operator surrogates
  - Cardinality invariant scalability



- **Challenges**
  - Mixed-integer decision domains
  - Gradients ill conditioning for deep graphs
  - Offline pre-training vs online adaptation
  - Computational cost of guarantees
- **Code**
  - github.com/SOLARIS-JHU/CINOC
  - github.com/Centrum-IntelliPhysics/PDEControl_DPC
  - github.com/pnnl/neuromancer
- **Hiring!**
  - Interested? Shoot me an email:

jdrgona1@jh.edu

NeuroMANCER Scientific Machine Learning Library



Open-source library in PyTorch

- Physics-informed Neural Networks
- Learning to optimize
- Neural differential equations
- Learning to control

github.com/pnnl/neuromancer

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DPC vs Model-based Reinforcement Learning

DPC is closely related to **MBRL** in that both leverage **model of dynamics**, but DPC utilizes **differentiable** closed-loop models and cost functions, allowing direct policy gradients **without requiring a learned critic**.

Empirical risk minimization problem:

$$\min_{\theta} \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\xi}_k} \left(\sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\xi}_k) + p_N(\mathbf{x}_N) \right)$$

$$\text{s.t. } \mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \boldsymbol{\xi}_k), \;\; k \in \mathbb{N}_0^{N-1}$$

$$\mathbf{u}_k = \pi_\theta(\mathbf{x}_k, \boldsymbol{\xi}_k)$$

$$h(\mathbf{x}_k, \boldsymbol{\xi}_k) \leq 0$$

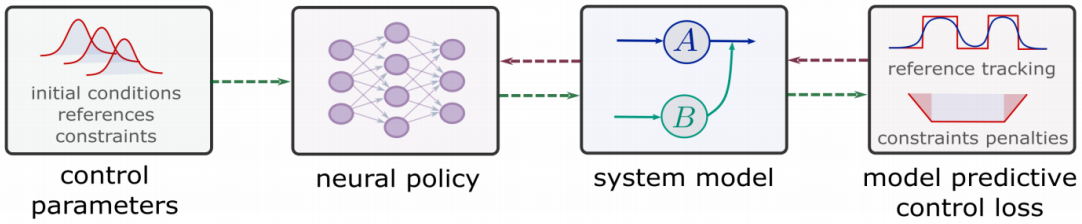
$$g(\mathbf{u}_k, \boldsymbol{\xi}_k) \leq 0$$

Critic parametrized by differentiable MPC loss function:

$$\mathcal{L}_{\text{DPC}} = \frac{1}{mN} \sum_{i=1}^{m} \left(\sum_{k=0}^{N-1} \left(\ell(\mathbf{x}_k^i, \mathbf{u}_k^i, \boldsymbol{\xi}_k^i) + p_x(h(\mathbf{x}_k^i, \boldsymbol{\xi}_k^i)) + p_u(g(\mathbf{u}_k^i, \boldsymbol{\xi}_k^i)) \right) + p_N(\mathbf{x}_N^i) \right)$$

Policy gradient via automatic differentiation:

$$\nabla_\theta \mathcal{L}_{\text{DPC}} = \left(\frac{\partial \ell}{\partial \mathbf{x}} + \frac{\partial p_x}{\partial \mathbf{x}} + \frac{\partial p_N}{\partial \mathbf{x}} \right) \cdot \frac{\partial \mathbf{x}}{\partial \mathbf{u}} \cdot \frac{\partial \mathbf{u}}{\partial \theta} + \left(\frac{\partial \ell}{\partial \mathbf{u}} + \frac{\partial p_u}{\partial \mathbf{u}} \right) \cdot \frac{\partial \mathbf{u}}{\partial \theta}$$



$\ell(\cdot)$: stage cost
$p_x(\cdot)$, $p_u(\cdot)$: constraints penalties
$p_N(\cdot)$: terminal cost
$\frac{\partial \mathbf{x}}{\partial \mathbf{u}}$: differentiable dynamics
$\frac{\partial \mathbf{u}}{\partial \theta}$: differentiable policy

J. Drgoňa, A. Tuor and D. Vrabie, *Learning Constrained Parametric Differentiable Predictive Control Policies With Guarantees, in IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024*

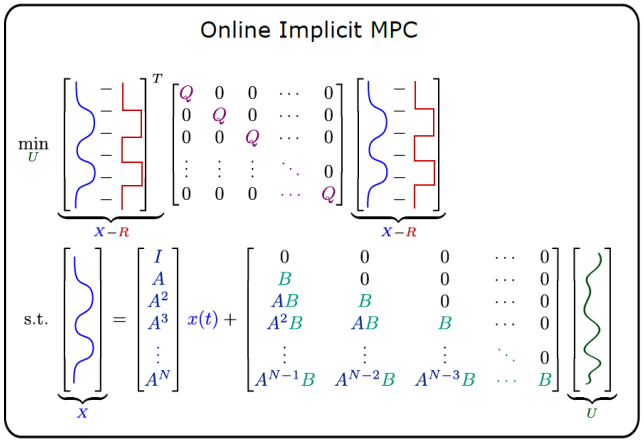
J. Drgoňa, et al., *Differentiable Predictive Control: An MPC Alternative for Unknown Nonlinear Systems using Constrained Deep Learning*, Journal of Process Control, 2022

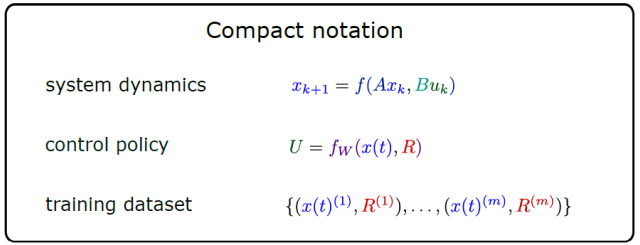
DPC vs Model Predictive Control

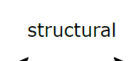
DPC is also closely related to MPC, but instead of single instance online optimization, the **DPC learns parametric explicit policy offline**, over distribution of parametric instances in batched setting.

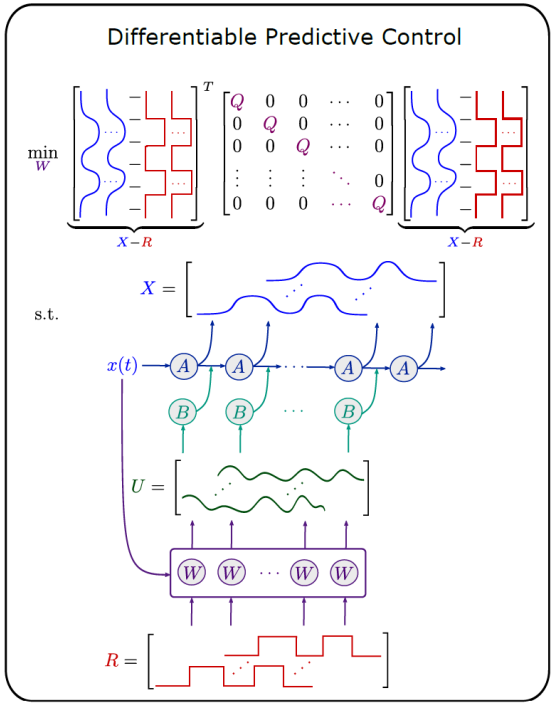
There is a **structural equivalence** between *single shooting* formulation of MPC problem and the *unrolled closed-loop system* dynamics in DPC.

DPC is also related to **explicit MPC**, as it solves a **parametric optimal control problem** via gradient-based policy optimization.









J. Drgoňa, A. Tuor and D. Vrabie, *Learning Constrained Parametric Differentiable Predictive Control Policies With Guarantees, in IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024*

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